

Sample Pages

From the Owner-Builder Job Site Binder — Second Edition, 2026

This free sample contains two complete documents exactly as they appear in the full binder:

1.5 Foundation Checklist — all 12 pages. Pre-pour preparation, pour-day execution, and post-pour curing, from batter boards to backfill.

8.2 Span Tables — all 5 pages. Floor joists, ceiling joists, rafters, headers, and deck joists, transcribed from the 2021 International Residential Code.

The full binder

367 pages across 8 sections — 57 print-ready PDFs plus editable Word contracts and auto-calculating Excel workbooks. One-time \$97, instant download, 30-day guarantee.

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1.5 Foundation Checklist

Pre-pour preparation, pour-day execution, and post-pour finishing — every step from batter boards to backfill.

Project Name: _____ Foundation Type: _____

Owner-Builder: _____ Start Date: _____

Foundation Contractor: _____ Target Completion: _____

INSTRUCTIONS: This checklist covers all phases of foundation construction: pre-pour preparation, pour day activities, and post-pour finishing. A proper foundation is critical — take your time, double-check measurements, and don't skip inspections. Document everything with photos.

⚠ CRITICAL REMINDER — Foundation work cannot proceed until:

- Building permit is approved and posted on site
- Site preparation is 100% complete
- All utilities are located (811 call completed)
- Foundation layout has been verified and approved

SECTION A: PRE-POUR PREPARATION

A1: Foundation Layout & Batter Boards

	Task	Date	Notes / Measurements
☐	Review foundation plans and verify all dimensions		
☐	Establish reference point/benchmark (permanent marker, stake, etc.)		
☐	Set up batter boards 3-4 feet outside all corners		
☐	String layout lines between batter boards		
☐	Verify all corners are square (3-4-5 method or diagonal measurement)		
☐	Diagonal measurements (must be equal!): A: _____ B: _____		

A1: Foundation Layout & Batter Boards

	Task	Date	Notes / Measurements
☐	Mark excavation limits with spray paint or stakes		
☐	Take photos of string layout from multiple angles		
☐	Have layout verified by surveyor or experienced builder (recommended)		

A2: Excavation

	Task	Date	Notes / Measurements
☐	Verify frost depth requirement (check local code): Frost depth (in): _____		
☐	Excavate to required depth below finish grade (frost depth + footer)		
☐	Verify excavation depth at all corners and mid-points (record below)		
☐	NE: _____ NW: _____ SE: _____ SW: _____		
☐	Remove all loose soil, roots, organic matter from excavation		
☐	Ensure excavation bottom is level and undisturbed soil		
☐	Document any rock, ledge, or unusual soil conditions encountered		
☐	Stockpile excavated soil away from work area (for later backfill)		

A3: Gravel Base & Compaction

	Task	Date	Notes / Measurements
☐	Install gravel base per plan (typically 4-6" of crushed stone)		
☐	Gravel type used: _____ Depth (in): _____		
☐	Compact gravel base with plate compactor or roller		

A3: Gravel Base & Compaction

	Task	Date	Notes / Measurements
☐	Verify base is level (max 1/4" variance in 10 feet)		
☐	Install drainage tile around perimeter (if required)		
☐	Verify drainage tile slopes to daylight or sump (1/8" per foot min)		

A4: Footer Forms & Rebar

	Task	Date	Notes / Measurements
☐	Install footer forms per plan dimensions		
☐	Footer width (in): _____ Footer depth (in): _____		
☐	Forms are straight, level, and properly braced		
☐	Verify footer width and depth at multiple locations		
☐	Install rebar per plan (typically #4 or #5, 2 bars continuous)		
☐	Rebar size: # _____ Configuration: _____		
☐	Rebar is elevated on chairs (3" min clearance from bottom)		
☐	All rebar splices overlap minimum 18" and are tied with wire		
☐	Rebar is clean (no rust scale, oil, or debris)		
☐	Corner rebar is properly bent or overlapped (no gaps)		

A5: Plumbing Under-Slab Rough-In

	Task	Date	Notes / Measurements
☐	Install all plumbing stubs per plan (water supply, drains, vents)		
☐	Verify plumbing stub locations match plan dimensions		

A5: Plumbing Under-Slab Rough-In

	Task	Date	Notes / Measurements
☐	All drain pipes slope properly (1/4" per foot minimum)		
☐	Pressure test water supply lines (if required by inspector)		
☐	Cap all plumbing stubs to prevent concrete entry during pour		
☐	Support all pipes properly (no sagging, proper bedding in gravel)		
☐	Take photos of all plumbing locations before pour		

A6: Vapor Barrier (for slab-on-grade)

	Task	Date	Notes / Measurements
☐	Install vapor barrier (minimum 6 mil poly, 10 mil preferred)		
☐	Overlap all seams minimum 6" and tape with approved sealing tape		
☐	Run vapor barrier up sides of forms (3-6" above slab height)		
☐	Seal vapor barrier around all plumbing penetrations		
☐	Repair any tears or punctures with approved patch/tape		
☐	Place sand layer over vapor barrier if required (prevents punctures)		

A7: Foundation Wall Forms (if applicable)

	Task	Date	Notes / Measurements
☐	Erect foundation wall forms (ICF, wood, or metal forms)		
☐	Wall height (in): _____ Wall thickness (in): _____		
☐	Forms are plumb, level, and properly braced (check with 4' level)		

A7: Foundation Wall Forms (if applicable)

	Task	Date	Notes / Measurements
☐	Install tie rods/snap ties at proper spacing per form manufacturer		
☐	Install vertical rebar in walls per plan (typically #4 @ 16-24" o.c.)		
☐	Install horizontal rebar per plan (min 2 continuous bars)		
☐	Tie vertical rebar to footer rebar (dowels or epoxy anchors)		
☐	Install anchor bolts at proper spacing (typically 6' o.c., 12" from corners)		
☐	Anchor bolt size: _____ Embedment depth (in): _____		
☐	Create forms for window wells, bulkhead openings (if applicable)		
☐	Coat forms with release agent/oil (prevents concrete adhesion)		
☐	Double-check all dimensions before calling for inspection		

A8: Pre-Inspection Checklist

Critical items to verify before calling the inspector:

- ☐ Foundation layout is square (diagonals match within 1/4")
- ☐ Excavation depth meets or exceeds frost depth requirement
- ☐ Footer dimensions (width and depth) match approved plans
- ☐ Rebar size, spacing, and placement meet code requirements
- ☐ Rebar has proper concrete cover (3" min from soil, 2" from forms)
- ☐ All plumbing under slab is installed and positioned correctly
- ☐ Vapor barrier is intact with no tears or unsealed penetrations
- ☐ Forms are clean, braced, and will not move during concrete pour
- ☐ Anchor bolts are on-site and ready to install during pour

- ☐ Site is accessible for inspector vehicle and concrete trucks
- ☐ Building permit is posted in visible location on site
- ☐ Approved foundation plans are available on-site for inspector review

A9: Foundation Inspection

⚠ **STOP — Do not pour concrete until inspection is passed.**

Inspection Requested: _____ Inspector: _____

Inspection Date: _____ Time: _____ Weather: _____

INSPECTION RESULT: ☐ PASSED ☐ FAILED ☐ CONDITIONAL

Inspector Comments / Requirements

- ☐ All inspector-required corrections completed and re-inspected (if needed)

Final Approval Date: _____ Approved to Pour (initials): _____

SECTION B: POUR DAY PREPARATION & EXECUTION**B1: Concrete Order & Scheduling**

	Task	Date	Notes
☐	Calculate concrete quantity needed (cubic yards) — add 5% waste		
☐	Quantity ordered (yd ³): _____ Mix design: _____		
☐	Specify concrete strength (typically 3000–4000 PSI for foundations)		
☐	Request appropriate slump (4–5" typical, 6–7" if using pump)		
☐	Order fiber reinforcement additive (if specified in plans)		

B1: Concrete Order & Scheduling

	Task	Date	Notes
☐	Ready-mix company: _____ Phone: _____		
☐	Pour date scheduled: _____ Arrival time: _____		
☐	Number of trucks: _____ Load spacing (min): _____		
☐	Confirm pump truck if needed: Company: _____ Time: _____		

B2: Weather & Site Conditions

	Task	Date	Notes
☐	Check 3-day weather forecast (no rain during pour, no freeze for 7 days)		
☐	Pour day forecast: High °F: _____ Low °F: _____ Precip? Y / N _____		
☐	Reschedule if temperature will be below 40°F or above 90°F		
☐	Have tarps/plastic sheeting ready in case of unexpected rain		
☐	Have insulated blankets ready if cold weather (below 50°F)		
☐	Verify access road is dry and firm enough for heavy trucks		

B3: Tools, Materials & Labor

Items needed on-site before the pour begins:

Tools & Equipment

- ☐ Wheelbarrows (at least 2) and shovels (multiple)
- ☐ Concrete vibrator (electric or gas-powered)

- ☐ Bull float and hand floats (magnesium or wood)
- ☐ Screed board (straight 2x4 or 2x6, longer than pour width)
- ☐ Hand tamper and concrete rake
- ☐ 4-foot level and tape measure
- ☐ Edging tools and grooving tools (for control joints)
- ☐ Rubber boots and gloves for all workers

Materials

- ☐ Anchor bolts (verify correct size/length, have extras)
- ☐ Curing compound or plastic sheeting for curing
- ☐ Water source for cleaning tools and curing concrete
- ☐ Form release oil (if not already applied)
- ☐ Foam backer rod for control joints (if applicable)

Labor

- ☐ Minimum 4–6 workers lined up and confirmed for pour day
- ☐ Workers briefed on safety (wet concrete is caustic — wear PPE!)
- ☐ Assign specific roles: screeding, floating, vibrating, tool cleaning
- ☐ First aid kit on-site with eye wash (concrete in eyes = emergency!)

B4: During the Pour

	Task	Date	Notes
☐	Pour start time: _____ Truck #1 arrival: _____		
☐	Verify concrete slump before discharge (3–6" typical)		
☐	DO NOT add water to concrete without reducing payment (weakens mix!)		
☐	Pour concrete in sections, working systematically (don't overfill areas)		
☐	Vibrate concrete thoroughly to eliminate voids (especially near forms)		

B4: During the Pour

	Task	Date	Notes
☐	DO NOT over-vibrate (causes segregation and weak concrete)		
☐	Monitor forms for movement, bulging, or failure during pour		
☐	Install anchor bolts while concrete is still workable		
☐	Verify anchor bolt spacing and alignment (check with string line)		
☐	Screed concrete level with top of forms (use straight 2x4)		
☐	Bull float surface to smooth and level		
☐	Create control joints if pouring slab (every 10-15 feet)		
☐	Take concrete test cylinders (if required by inspector/engineer)		
☐	Pour completion time: _____ Total duration (hrs): _____		
☐	Total concrete used (yd ³): _____ Waste/short (yd ³): _____		

SECTION C: POST-POUR FINISHING & CURING**C1: Finishing (if slab/exposed surface)**

	Task	Date	Notes
☐	Wait for bleed water to evaporate (do not finish while water is present!)		
☐	Hand float surface to desired texture		
☐	Edge all perimeter areas with edging tool		
☐	Groove control joints to 1/4 depth of slab		

C1: Finishing (if slab/exposed surface)

	Task	Date	Notes
☐	Broom finish surface if non-slip texture desired		
☐	Do final troweling when concrete is firm enough to support weight		

C2: Curing

	Task	Date	Notes
☐	Apply curing compound OR cover with plastic sheeting (within 12 hours)		
☐	Keep concrete moist for minimum 7 days (14 days in cold weather)		
☐	Protect from freezing temperatures (cover with insulated blankets)		
☐	Protect from excessive heat and sun (mist with water if needed)		
☐	Protect from rain damage in first 24 hours (cover with tarps)		

C3: Form Removal & Waterproofing

	Task	Date	Notes
☐	Wait minimum 3 days before removing forms (7 days in cold weather)		
☐	Remove forms carefully to avoid chipping or damaging concrete		
☐	Remove snap ties and patch holes with mortar/hydraulic cement		
☐	Inspect foundation walls for honeycomb, voids, or defects		
☐	Repair any defects before waterproofing (surface must be sound)		
☐	Apply dampproofing/waterproofing per plan (tar, membrane, etc.)		
☐	Install drainage board or protection board over waterproofing		
☐	Verify perimeter drain tile is in place and functional		

C4: Backfill & Compaction

	Task	Date	Notes
☐	Wait minimum 7 days after pour before backfilling (14 days preferred)		
☐	Ensure waterproofing is fully cured before backfill		
☐	Remove large rocks and debris from backfill material		
☐	Backfill in 12" lifts, compacting each layer (plate compactor)		
☐	Keep heavy equipment away from walls (compact by hand near walls)		
☐	Grade soil away from foundation (2% min slope, 6" drop in 10 feet)		
☐	Protect foundation from erosion until final grading/landscaping		

C5: Final Inspection & Approval

Final Inspection Requested: _____ Inspector: _____

Inspection Date: _____ Time: _____

- ☐ Foundation is fully cured (minimum 7 days since pour)
- ☐ Forms removed and all repairs completed
- ☐ Waterproofing applied and drainage installed
- ☐ Backfill completed and properly compacted
- ☐ Final grading slopes away from foundation

FINAL INSPECTION RESULT: ☐ PASSED ☐ FAILED ☐ CONDITIONAL

Inspector Comments

Approved to Proceed with Framing (initials): _____ Date: _____

FOUNDATION COMPLETION SIGN-OFF

I certify that the foundation has been constructed according to approved plans and all inspections have passed. The foundation is ready for the next phase of construction.

Owner-Builder Signature: _____ Date: _____

Foundation Contractor (if applicable): _____ Date: _____

PHOTO DOCUMENTATION — FOUNDATION

Document all critical stages with photos. These are essential for warranty claims, inspection appeals, and future reference.

Date	Photo Description	File Name / Number	Notes
	Excavation complete and depth verified		
	Rebar installed before inspection		
	Plumbing under-slab installed		
	Vapor barrier and gravel base		
	Forms ready for inspection		
	During pour — concrete placement		
	Anchor bolts installed		
	Finished surface after pour		
	Forms removed — walls complete		
	Waterproofing applied		
	Drainage system installed		
	Backfill completed		

NOTES, ISSUES & LESSONS LEARNED

8.2 Span Tables

Maximum allowable spans for No. 2 grade dimensional lumber, from the 2021 International Residential Code. Values are maximums — your plans and local amendments govern.

Before you use these tables

- Values assume No. 2 grade lumber, single span, bearing at each end. Spans are in feet-inches.
- Your local jurisdiction may amend these values. The approved plans and your building department always govern.
- Engineered lumber (I-joists, LVL) has different span capabilities — use the manufacturer's tables.

FLOOR JOISTS — Living Areas

40 psf live load, 10 psf dead load, deflection limit L/360. IRC Table R502.3.1(2).

Floor Joists — 2x8

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	13'-6"	11'-10"	10'-10"	9'-8"
Douglas Fir-Larch	14'-2"	12'-9"	11'-8"	10'-5"
Hem-Fir	13'-2"	12'-0"	11'-3"	10'-2"
Spruce-Pine-Fir	13'-6"	12'-3"	11'-8"	10'-5"

Floor Joists — 2x10

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	16'-2"	14'-0"	12'-10"	11'-5"
Douglas Fir-Larch	18'-0"	15'-7"	14'-3"	12'-9"
Hem-Fir	16'-10"	15'-2"	13'-10"	12'-5"
Spruce-Pine-Fir	17'-3"	15'-5"	14'-1"	12'-7"

Floor Joists — 2x12

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	19'-1"	16'-6"	15'-1"	13'-6"

Floor Joists — 2x12

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Douglas Fir-Larch	20'-11"	18'-1"	16'-6"	14'-9"
Hem-Fir	20'-4"	17'-7"	16'-1"	14'-4"
Spruce-Pine-Fir	20'-7"	17'-10"	16'-3"	14'-7"

Notes: O.C. = on center. For cantilevers, maximum overhang = 1/4 of the back span. Provide blocking or bridging at mid-span for joists over 8 feet.

CEILING JOISTS — Attic With Limited Storage

20 psf live load (uninhabitable attic with limited storage), deflection limit L/240. IRC Table R802.4.1(2). If your attic will never carry storage, code allows longer spans — these values are the safe, conservative choice.

Ceiling Joists — 2x6

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	13'-11"	12'-0"	11'-0"	9'-10"
Douglas Fir-Larch	15'-0"	13'-0"	11'-11"	10'-8"
Hem-Fir	14'-5"	12'-8"	11'-7"	10'-4"
Spruce-Pine-Fir	14'-9"	12'-10"	11'-9"	10'-6"

Ceiling Joists — 2x8

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	17'-7"	15'-3"	13'-11"	12'-6"
Douglas Fir-Larch	19'-1"	16'-6"	15'-1"	13'-6"
Hem-Fir	18'-6"	16'-0"	14'-8"	13'-1"
Spruce-Pine-Fir	18'-9"	16'-3"	14'-10"	13'-3"

Ceiling Joists — 2x10

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	20'-11"	18'-1"	16'-6"	14'-9"
Douglas Fir-Larch	23'-3"	20'-2"	18'-5"	16'-5"
Hem-Fir	22'-7"	19'-7"	17'-10"	16'-0"
Spruce-Pine-Fir	22'-11"	19'-10"	18'-2"	16'-3"

COMMON RAFTERS — Snow Load Areas

30 psf ground snow load, 10 psf dead load, ceiling not attached to rafters, deflection limit L/180. IRC 2021 Table R802.4.1(3). Higher snow areas require shorter spans — check your local ground snow load.

Rafters — 2x6

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	12'-11"	11'-2"	10'-2"	9'-2"
Douglas Fir-Larch	14'-0"	12'-1"	11'-0"	9'-10"
Hem-Fir	13'-7"	11'-9"	10'-9"	9'-7"
Spruce-Pine-Fir	13'-9"	11'-11"	10'-11"	9'-9"

Rafters — 2x8

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	16'-4"	14'-2"	12'-11"	11'-7"
Douglas Fir-Larch	17'-8"	15'-4"	14'-0"	12'-6"
Hem-Fir	17'-2"	14'-11"	13'-7"	12'-2"
Spruce-Pine-Fir	17'-5"	15'-1"	13'-9"	12'-4"

Rafters — 2x10

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	19'-5"	16'-10"	15'-4"	13'-9"
Douglas Fir-Larch	21'-7"	18'-9"	17'-1"	15'-3"
Hem-Fir	21'-0"	18'-2"	16'-7"	14'-10"
Spruce-Pine-Fir	21'-4"	18'-5"	16'-10"	15'-1"

Rafters — 2x12

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	22'-10"	19'-10"	18'-1"	16'-2"
Douglas Fir-Larch	25'-1"	21'-8"	19'-10"	17'-9"
Hem-Fir	24'-4"	21'-1"	19'-3"	17'-3"
Spruce-Pine-Fir	24'-8"	21'-5"	19'-6"	17'-6"

HEADERS & GIRDERS — Exterior Bearing Walls

Douglas Fir-Larch, Hem-Fir, Southern Pine, or Spruce-Pine-Fir, No. 2 or better. 30 psf ground snow, 24-foot building width. IRC 2021 Table R602.7(1). "Jacks" = number of jack (trimmer) studs required at EACH end of the header.

Header Spans (span + jack studs per end)

Header Size	Supporting Roof + Ceiling	+ One Floor	+ Two Floors
2-2x4	3'-1" (1 jack)	2'-6" (1 jack)	2'-1" (1 jack)
2-2x6	4'-7" (1 jack)	3'-9" (1 jack)	3'-2" (2 jacks)
2-2x8	5'-9" (1 jack)	4'-10" (2 jacks)	4'-0" (2 jacks)
2-2x10	6'-10" (2 jacks)	5'-8" (2 jacks)	4'-9" (2 jacks)
2-2x12	8'-1" (2 jacks)	6'-8" (2 jacks)	5'-7" (2 jacks)
3-2x8	7'-3" (1 jack)	6'-0" (1 jack)	5'-0" (2 jacks)
3-2x10	8'-7" (1 jack)	7'-2" (2 jacks)	5'-11" (2 jacks)
3-2x12	10'-1" (2 jacks)	8'-5" (2 jacks)	7'-0" (2 jacks)
4-2x8	8'-4" (1 jack)	6'-11" (1 jack)	5'-9" (1 jack)
4-2x10	9'-11" (1 jack)	8'-3" (2 jacks)	6'-10" (2 jacks)
4-2x12	11'-8" (1 jack)	9'-8" (2 jacks)	8'-1" (2 jacks)

Notes: "+ One Floor" / "+ Two Floors" = header also supports one or two center-bearing floors above. Install headers on edge, crown up. King stud required each side in addition to jack studs. Wider buildings, higher snow loads, or clear-span floors above require larger headers — see IRC Table R602.7(1) or consult an engineer.

DECK JOISTS

40 psf live load, 10 psf dead load, wet service. Span measured from face of support to face of support, no overhang. IRC 2021 Table R507.6.

Deck Joists — Southern Pine #2

Joist Size	12" O.C.	16" O.C.	24" O.C.
2x6	9'-11"	9'-0"	7'-7"
2x8	13'-1"	11'-10"	9'-8"
2x10	16'-2"	14'-0"	11'-5"
2x12	18'-0"	16'-6"	13'-4"

Deck Joists — Doug Fir-Larch / Hem-Fir / SPF #2

Joist Size	12" O.C.	16" O.C.	24" O.C.
2x6	8'-10"	8'-0"	6'-10"
2x8	11'-8"	10'-7"	8'-8"
2x10	14'-11"	13'-0"	10'-7"
2x12	17'-5"	15'-1"	12'-4"

Deck notes: use pressure-treated or naturally durable lumber; joist hangers at ledger connections; cantilever (overhang) limited to 1/4 of the actual joist span; check local amendments.

IMPORTANT DISCLAIMER

These tables are general reference for common residential conditions, transcribed from the 2021 International Residential Code. Actual requirements vary with species, grade, moisture, load duration, snow load, and local amendments. Confirm all structural sizing with your building department and, where required, a licensed engineer.